groundwater to water resources used by man; this is schematically illustrated in Figure 22.1. In particular we discuss actinide behavior as these elements have the most hazardous radionuclides which may be released in the different steps of the nuclear fuel cycle, and, especially, from nuclear waste repositories. However, despite this it may be noted here that the actinides contribution to the dose from a leaking cannister to the recipient is negligible. This is exemplified by the dose to a recipient from one leaking cannsiter in the planned Swedish repository shown in Figure 22.2.

22.1. RADIOACTIVE RELEASES AND POSSIBLE EFFECTS

In earlier chapters, there have been brief discussions of the release of radionuclides to the environment. Such releases occur from mining and milling operations, particularly of uranium ores (§14.2.4.4), from the nuclear fuel fabrication processes, from normal operation of nuclear reactors (§19.16), from reprocessing of spent nuclear fuel (§21.8), from nuclear weapons production and recovery, from transportation of nuclear material, from testing of nuclear weapons and accidents (§13.10), and from storage of nuclear wastes (which is discussed in subsequent sections). Table 22.1 compares the estimated collective dose to the public from the various activities of the nuclear fuel cycle, from

Table 22.1 Collective effective dose to the public from radionuclides released in effluents from the nuclear fuel cycle 1995-97 (UNSCEAR 2000)

Source	Normalized collective effective dose (manSv/GW_ey)
	Local and regional effects
Mining	0.19
Milling	0.008
Mine and mill tailings (releases over 5 years)	0.04
Fuel fabrication	0.003
Reactor operation	
Atmospheric	0.4
Aquatic	0.04
Reprocessing	
Atmospheric	0.04
Aquatic	0.09
Transportation	<0.1
	Solid waste disposal and global effects
Mine and mill tailings	
Releases of radon over 10 000 y	7.5
Reactor operation	
Low-level waste disposal	0.00005
Intermediate-level waste disposal	0.5
Reprocessing solid waste disposal	0.05
Globally dispersed radionuclides (truncated to 10 000 years)	40

Table 22.2 Radionuclides released in atmospheric nuclear testing (UNSCEAR2000)

Radio-nuclide	Half-life	Estimated release Total (EBq)	Radio-nuclide	Half life	Estimated release Total (EBq)
^{3}H	12.33 y	186	^{125}Sb	2.73 y	0.741
$^{14}C^{\dagger}$	5730. y	0.213	^{131}I	8.02 d	675
^{54}Mn	312.3 d	3.98	^{137}Cs	30.07 y	0.948
^{55}Fe	2.73 y	1.53	^{140}Ba	12.75 d	759
^{89}Sr	50.53 d	117	^{141}Ce	32.50 d	263
^{90}Sr	28.78 y	0.622	^{144}Ce	284.9 d	30.7
^{91}Y	58.51 d	120	^{239}Pu	24110. y	0.00652
^{95}Zr	64.02 d	148	^{240}Pu	6560. y	0.00435
^{103}Ru	39.26 d	247	^{241}Pu	14.36 y	0.142
^{106}Ru	1.023 y	12.2			

For the non-gaseous fission products a total non-local fission explosion yield of 160.5 Mt, obtained from measured ^{90}Sr deposition, was assumed in deriving the total amounts released.
†For simplicity, all ^{14}C is assumed to be due to fusion.

mining and milling through nuclear waste disposal; the significant point is the recognition that the mining operation for uranium is the major contributor in the dose to the global public by quite a large fraction.

An estimate of the total radioactivity, by nuclide, released into the atmosphere by above-ground nuclear tests is given in Table 22.2; the total release is ∼2 600 EBq. To provide comparison, the estimated global releases from reactors and reprocessing plants are listed in Table 22.3; globally this gives ∼3.9 EBq from reactors and 3.4 EBq from reprocessing plants up to 1998. The major releases are the noble gases, ^{3}H and ^{14}C. However, the total ^{14}C release over 30 years is less than one percent of the normal ^{14}C level from cosmic ray production (§ 13.4.3). This is a small fraction of the radioactivity released in nuclear weapons testing. To further place the releases in proper relation, an estimate of the total atmospheric release in the Chernobyl accident was 2 EBq (Table 22.4), in the Three Mile Island Reactor (TMI) accident, 10^{-6} EBq, and from the Fukushima accident ∼0.9 EBq. From such figures the United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) estimates that the average annual dose of radiation per person during year 2000 corresponded to:

Natural background	2.4 mSv
Medical diagnostics	0.4 mSv
Nuclear weapons tests	0.005 mSv
Chernobyl accident	0.002 mSv
Nuclear fuel cycle	0.0002 mSv

Table 22.3 Global release of radionuclides by reactors and reprocessing plants up to 1998 (UNSCEAR2000)

Radionuclides	Reactor releases (PBq)	Reprocessing releases (PBq)
Noble gases	3631	3190
^{3}H	269	144
^{14}C	1.97	0.44
^{90}Sr	–	6.6
^{106}Ru	–	19
^{129}I	–	0.014
^{131}I	0.046	0.004
^{137}Cs	–	40
Particulates	0.121	–
Others	0.839	–
Total	~3900	~3400

Table 22.4 Radionuclides released into the atmosphere in the Chernobyl accident and local contamination

		Release		Core inventory tot EBq	Contamination	
Nuclide	Half-life	Tot Ebq	%		Air (Bq m ^{3}) Stockholm †	Ground (kBq m ^{2}) Gävle area ‡
^{85}Kr	10.73 y	0.033	100	0.033		
^{89}Sr	50.5 d	0.094	4.0	2.4		
^{90}Sr	28.6 y	0.0081	4.0	0.20		
^{95}Zr	64.0 d	0.16	3.2	5.0	0.6	5.9
^{103}Ru	39.4 d	0.14	2.9	4.8	6.4	14.6
^{106}Ru	368 d	0.059	2.9	2.0	1.8	4.0
^{131}I	8.04 d	0.67	20	3.3	15	179
^{133}Xe	5.24 d	1.7	≈100	1.7		
^{134}Cs	2.07 y	0.019	10	0.19	2.4	14.4
^{136}Cs	13.2 d				0.7	6.0
^{137}Cs	30.2 y	0.037	13	0.28	4.5	24.7
^{140}Ba	12.8 d	0.28	5.6	5.0	25.6	2.1
^{141}Ce	32.5 d	0.13	2.3	5.6	0.5	5.9
^{144}Ce	284 d	0.088	2.8	3.1	0.3	3.7
^{239}Np	2.36 d	0.97	3.2	30		2.7
^{238}Pu	87.7 y	3.0×10^{-5}	3	0.001		
^{239}Pu	24100 y	2.6×10^{-5}	3	0.0009		
^{240}Pu	6570 y	3.7×10^{-5}	3	0.0012		
^{241}Pu	14.4 y			0.170		

†First days, April 28-29, 1986.
‡Ullbolsta, outside of Gävle, among highest depositions outside Russia; corrected to April 28, 1986. References: USSR State Comm. on the Utilization of Atomic Energy 1986. The accident at the Chernobyl nuclear power plant and its consequences, IAEA expert meeting 25-29 Aug. 1986, Vienna.

22.2. RADIONUCLIDES OF ENVIRONMENTAL CONCERN

Most of the radionuclides produced in nuclear tests, accidents and in the normal fuel cycle are short lived. In Table 21.3, longer lived fission products and activation products from these systems are listed as these represent the major concern to the general public if they are allowed to enter the environment; we exclude nuclides with insignificant contribution to the total activity after 10 years. In addition, several heavy radionuclides are formed by neutron capture reactions such as ^{236}U, ^{237}Np, $^{238-242}$Pu, ^{241}Am, etc.

To assess the potential for these radionuclides to cause harm to humans, their geochemical and biological behavior must be evaluated. For example, since Kr is a chemically inert gaseous element, it would have little effect on a person who inhaled and immediately exhaled the small amount which might be present in the air. By contrast, other nuclides with high activity, ^{90}Sr–^{90}Y and ^{137}Cs–^{137m}Ba, have active geological and biological behavior and can present much more significant radiation concerns to humans. In normal operations, these nuclides would of course be released in quite insignificant amounts, as represented by the value for aquatic releases in reactor operation and reprocessing, Table 19.7 and 21.10. Similarly, the heavy elements (Np, Pu, Am) would not be released in normal operation at levels that would be of concern. We have pointed out that some low level releases as a result of normal operations are allowed by the health authorities, who also monitor these levels. In the effluents from reprocessing plants (e.g. Sellafield in the United Kingdom and La Hague in France), the relatively long-lived nuclides such as ^{3}H, ^{14}C, ^{85}Kr, ^{99}Tc and ^{129}I are of major concern. The liquid effluents from nuclear power plants and from reprocessing plants are about equally responsible for the global collective dose commitment of nuclear power generation (i.e., $\sim$0.8 man Sv per GW_ey of the total $\sim$2.5 man Sv).

22.3. RELEASES FROM LARGE REACTOR ACCIDENTS

So far there have only been two major nuclear reactor accidents, Chernobyl and Fukushima. In both cases large amounts of radioactive material was released into the environment. However, in these events there was only an insignificant addition of ^{14}C to the atmosphere relative to the average atmospheric amount of ^{14}C at the corresponding times.

22.3.1. Chernobyl

On April 26, 1986, a low power engineering test was being conducted at one of the reactors of the Chernobyl nuclear power station in the Ukraine (then the USSR). The reactor became unstable, resulting in thermal explosions and fires that caused